

GUOHUI LI

Undergraduate Student in Information and Computing Science | UESTC
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EDUCATION

University of Electronic Science and Technology of China

B.S. Candidate in Information and Computing Science, School of Mathematical Sciences

- Ranked 1st in the major; recipient of the National Scholarship.

Chengdu, China

Sep. 2024 - Present

RESEARCH INTERESTS

Efficient and reliable multimodal large language models; multimodal retrieval-augmented generation; adaptive context compression; efficient multimodal reasoning.

PUBLICATIONS

LLaVA-FA: Learning Fourier Approximation for Compressing Large Multimodal Models

Pengcheng Zheng, Chaoning Zhang, Jiarong Mo, **GuoHui Li**, et al. | *ICLR 2026, Poster*

- Contributed to the design of joint Fourier-domain low-rank decomposition and quantization, and completed part of the theoretical derivation and method analysis.

TRACE: Topology-Aware Retrieval via Aspect Connectivity Estimation

Pengcheng Zheng*, **GuoHui Li***, Chaoning Zhang, et al. | *Under review at IEEE TKDE* | * Equal contribution

- Proposed and developed the core topology-aware reranking approach under the guidance of senior collaborators; implemented the framework and conducted the main experiments, ablation studies, and result analyses on three multi-hop QA benchmarks.

RESEARCH EXPERIENCE

Future Media Research Center, UESTC

Undergraduate Researcher, supervised by Prof. Chaoning Zhang

Since Apr. 25, 2025

- Conduct research on multimodal model compression and retrieval-augmented generation, including model implementation, evaluation, and experimental analysis.

SELECTED PROJECTS

Meow Loop - Unity Card Game

- Developed a light-strategy card game in Unity/C# for Tencent's "Sanjiao Cup"; contributed to gameplay design, programming, asset integration, and final delivery.

Arena Cleanup - Unreal Engine 5 Cooperative Shooter

- Implemented a UE5.7 cooperative shooter with wave-based enemy AI, direct-IP LAN multiplayer, synchronized game state, and Windows/Android builds.

HONORS AND AWARDS

National Scholarship | Provincial First Prize, Lanqiao Cup Software Competition | Honorable Mention, Mathematical Contest in Modeling | Provincial Second Prize, Chinese Mathematics Competitions

TECHNICAL SKILLS

Programming Languages: Python, C++, C#

Machine Learning and Retrieval: PyTorch, Hugging Face Transformers, Sentence-Transformers, FAISS

Tools: NumPy, scikit-learn, Linux, Git, LaTeX